

Candidate's index number:

**THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MOSHI DISTRICT COUNCIL
FORM ONE TERMINAL EXAMINATION
PHYSICS**

CODE NO.031

TIME: 02:30HRS

MAY, 2026

INSTRUCTIONS

1. This paper consists of ten (10) questions.
2. Answer all questions.
3. All work done and answers of each question must be shown clearly in the space provided.
4. Communication devices and any unauthorized materials are not allowed in the examination room.
5. Non-programmable calculators and mathematical tables may be used.
6. Write your **Index number** at the top right corner in the space provided.

FOR EXAMINER'S ONLY		
QUESTION	SCORE	INITIALS
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		

GEPAM science hub
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SECTION A: 15 Marks

Answer all questions in this section

1. For each of the items (i)-(x), choose the correct answer from the given alternatives and write its letter besides the item number in the space provided.
- (i) Physics is a fundamental branch of science because the knowledge obtained in studying physics is strongly connected to other branches of science. Which among the following statements describes how physics is connected to other disciplines like chemistry and biology?
- A. Physics studies the properties of matter and energy which are fundamental to all sciences
 - B. Physics focuses solely on celestial bodies and space exploration
 - C. Physics deals with the human mind and cognitive process
 - D. Physics have many applications in medical field
- (ii) Which of the following best describes a variable?
- A. A trend that show an experimental relationship
 - B. Something whose value can change over multiple measurements
 - C. A measure of how much a plot line changes along the y-axis
 - D. Something that remains constant over multiple measurements
- (iii) A cuboid with length 4cm , width 5cm and height 10 cm is made from wood. What is the actual volume of the cuboid?
- A. 200cm^3 .
 - B. 120cm^3 .
 - C. 240ml.
 - D. 120cm^3 .
- (iv) Which of the following is true if the mass of an object is reduced
- A. State of an object changes
 - B. Amount of matter is reduced
 - C. Volume of an object is reduced
 - D. Density of an object increases
- (v) Which instrument will you use to measure accurately the inside diameter of a bottle neck?
- A. Micrometer screw gauge
 - B. Meter rule
 - C. Vernier caliper
 - D. Tape measure
- (vi) A cylindrical container has a diameter of 10 cm and a height of 12 cm . the actual volume of this cylinder is
- A. 942 cm^3
 - B. 120 cm^3
 - C. 22 cm^3
 - D. 120 cm^3
- (vii) to measure the length using a meter rule, wrong position of the eye leads

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- A. parallax error
B. eye error
C. zero error
D. meter error
- (viii) For the aims of physics to be attained, physicists have divided physics into a number of branches. The following set represents branches of physics except.
A. Biophysics, geophysics, chemophysics and astronophysics
B. Heat, optics, mechanics and electronics
C. Wave, physics of the atom and physikos
D. Agribusiness, mechanics and molecules
- (ix). Astronomy deals with the study of the universe and everything contained in it. The word astronomy comes from two greek words which are:
A. Astro and star
B. Nomos and star
C. Astron and nomos.
D. Laws and star
- (x). Physics involves the study of matter energy and their mutual interaction. What is the advantage of the interaction between matter and energy in physics
A. It helps to explain the behaviour of objects and phenomena in the universe
B. It allows us to explore different planets and their atmosphere
C. It is not relevant to the study of physics
D. It helps to understand human body and its function

ANSWERS

Item	i	ii	iii	iv	v	vi	vii	viii	ix	x
Answer										

2. Match the item in list A with the correct responses from list B by writing the letter of the correct answer in the space provided.

List A	List B
I. Provided the elastic limit is not exceeded, the extension is directly proportional to the force applied	A. Ohms law B. Law of floatation C. Hooke's law D. Law of conservation of energy E. Archimedes' principle
II. A floating body displaces its own weight of fluid in which it floats	
III. Any object partially or totally immersed in a fluid experiences an upthrust which is	

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IV.	equal to the weight of the fluid displaced by the object	F. Pascal principle
V.	When pressure is applied at any point on the surface of a fluid contained in a closed container, the pressure is transmitted undiminished to all parts of the fluid and to the walls of the container	G. Law of motion
	Energy can neither be created nor destroyed but it can be transformed from one form to another	

LIST A	I	II	III	IV	V
LIST B					

SECTION B: 70 Marks

Answer all questions in this section

3. (a) what is physics?

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B) Discoveries in physics has led to various inventions that influence our lives. Briefly explain how physics is applied in the following fields

(i) At home:.....

.....

(ii) In hospitals

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(b) Scientists including physicists are always looking for scientific evidence. As a physics expert assist a form one student to outline the steps followed when carrying out a scientific investigation.

I.

II.

III.

IV.

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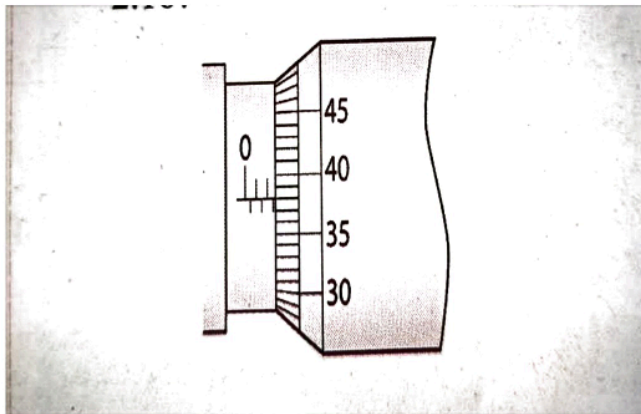
VI.

VII.

4. (a) Micrometer screw gauge usually used to measure the diameters of thin objects like wires and ball bearings. Briefly explain why we use Micrometer screw gauge instead of vernier caliper?

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- (b). What is the reading of a Micrometer screw gauge from the figure below



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5. (a) Physics relates to many subjects in terms of either direct applications or the principles governing the working of instruments. Explain how physics is related to the following subjects

(i) Chemistry :

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(ii) Biology.

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- (b) Sources of energy contribute to the development of modern society batteries, generators and solar cells are good examples of these source of energy what are the contributions of the sources of energy in our daily life

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6. (a) What is scientific theory:

(b) Theory of gravity explains the force of attraction between two objects that have mass. For our daily activities this force is very important. Mention four significance of the theory of gravity

I.

II.

III.

IV.

7. (a) With examples give the meaning of measurement of a physical quantity

(b) Draw a well labelled diagram of a micrometer screw gauge

8. (a) A form one student uses a vernier caliper to measure the internal diameter of a 100ml measuring cylinder. What are the steps used by this student to get the readings?

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(b) Errors usually arise due to several reasons. Briefly explain the three common errors caused by improper reading of an instrument

- I.
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- II.
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-
- III.
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9. (a) Why volume is said to be a derived quantity?

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(b) Calculate the volume of a rectangular block of sides 15 cm, 8cm and 7 cm

(c) Measuring the volume of an irregular shaped solid object is based on the archimedes principle which state that when an object is completely submerged in water, it displaces a volume of water equal to its own volume, the apparatus that can be used to measure volume of an irregular object are:

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SECTION C: 15 Marks

Answer question number ten (10)

10. (a) Differentiate between mass and volume of a substance (give three points)

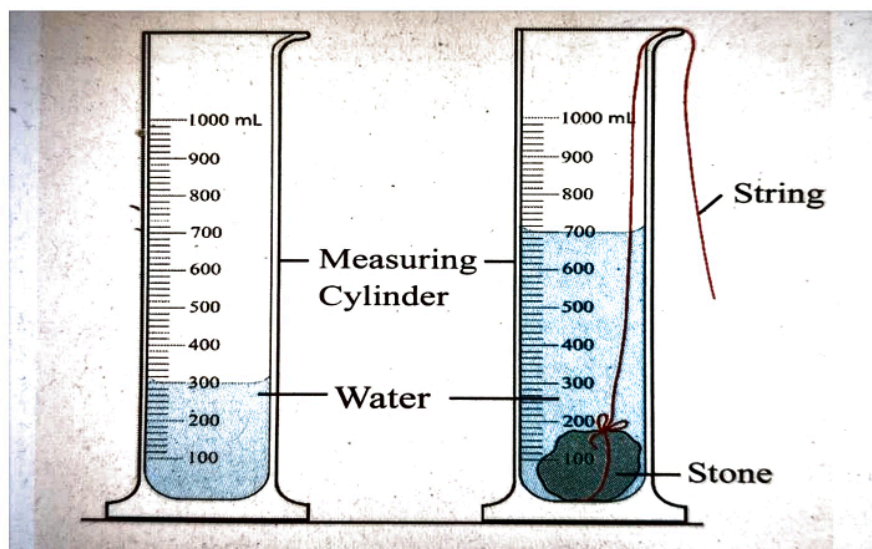
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(b) A figure below shows a measuring cylinder containing water before and after a stone is immersed. what is the volume of the stone?



(c) A rectangular block measures height 1.00cm, width 2.50 cm and length 4.00cm

(i) What instrument was used to measure the sides of the rectangular block?

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(ii) Calculate the volume of the rectangular block

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